

Offline coloring and indexing pipeline

Fixed sparse SMT
snapshot

Extract active nodes
and sibling intervals

Build laminar interval
forest and exact width m

First-fit feasible
exact-width coloring

ACTIVEBALANCE
subtree permutations

Balanced color stores
 $\mathcal{D}_c, \mathcal{M}_c$

Fixed-shape batch retrieval and proof completion

Client chooses
target leaf ℓ

Use interval metadata
to find one target per color

Send one PIR query
per color, dummy if needed

Write real outputs
into proof slots

Fill unmatched levels
with default hashes

Hash upward
to the root